Import Libraries

In [165…
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [127…
```
df = pd.read_csv(r"C:\Users\Siddhesh\Downloads\Teen_Mental_Health_Dataset (1).
```

In [129…
```
df.head()
```

Out[129…

	age	gender	daily_social_media_hours	platform_usage	sleep_hours	screen_t
0	14	male	7.9	Instagram	7.4	
1	19	female	1.9	TikTok	8.0	
2	17	female	1.3	Instagram	7.6	
3	15	male	7.4	TikTok	6.9	
4	15	female	4.7	Both	4.9	

In [131…
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 13 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   age                        1200 non-null   int64
 1   gender                     1200 non-null   object
 2   daily_social_media_hours   1200 non-null   float64
 3   platform_usage             1200 non-null   object
 4   sleep_hours                1200 non-null   float64
 5   screen_time_before_sleep   1200 non-null   float64
 6   academic_performance       1200 non-null   float64
 7   physical_activity          1200 non-null   float64
 8   social_interaction_level   1200 non-null   object
 9   stress_level               1200 non-null   int64
 10  anxiety_level              1200 non-null   int64
 11  addiction_level            1200 non-null   int64
 12  depression_label           1200 non-null   int64
dtypes: float64(5), int64(5), object(3)
memory usage: 122.0+ KB
```

In [17]:
```
df.describe()
```

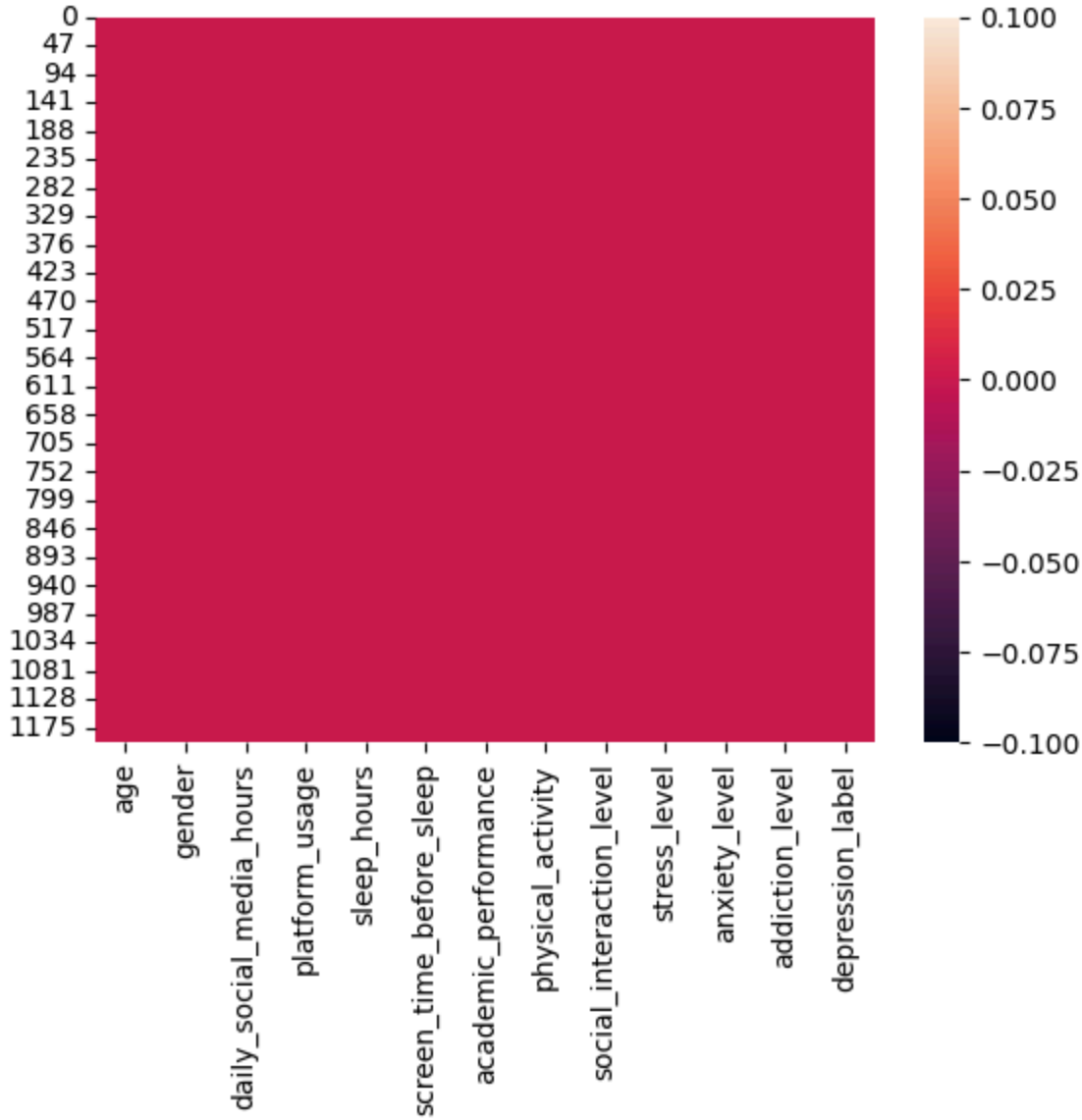
Out[17]:

	age	daily_social_media_hours	sleep_hours	screen_time_before_s
count	1200.000000	1200.000000	1200.000000	1200.00
mean	15.928333	4.536667	6.449417	1.74
std	2.021947	2.029599	1.442677	0.71
min	13.000000	1.000000	4.000000	0.50
25%	14.000000	2.800000	5.200000	1.10
50%	16.000000	4.500000	6.500000	1.80
75%	18.000000	6.300000	7.600000	2.40
max	19.000000	8.000000	9.000000	3.00

In [87]:
```
df.isnull().sum()
```

Out[87]:
```
age                        0
gender                     0
daily_social_media_hours   0
platform_usage             0
sleep_hours                0
screen_time_before_sleep   0
academic_performance       0
physical_activity          0
social_interaction_level   0
stress_level               0
anxiety_level              0
addiction_level            0
depression_label           0
screen_exposure            0
dtype: int64
```

In [21]:
```
sns.heatmap(df.isnull())
plt.show()
```



In [157…

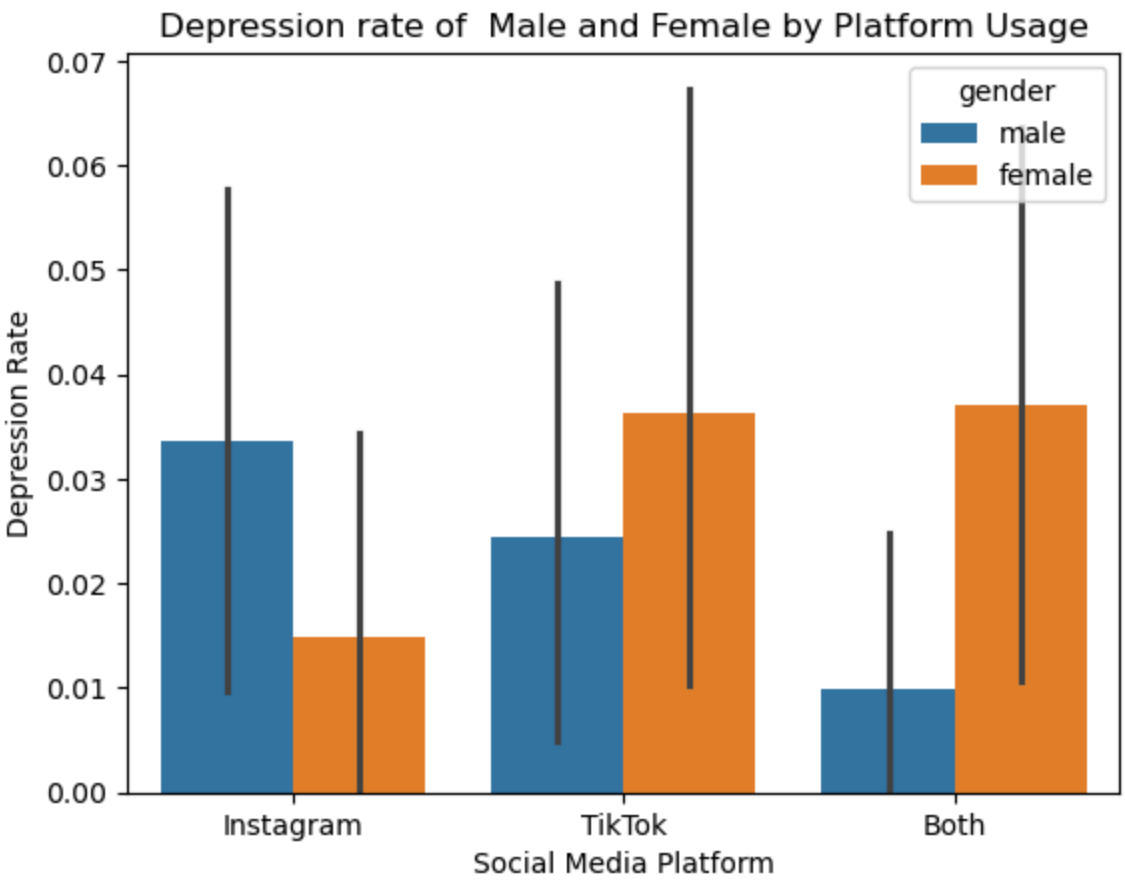
```
df['screen_exposure'] = df['sleep_hours'] + df['screen_time_before_sleep']
```

Performing EDA

Depression Rate of Gender vs Platform

In [137…

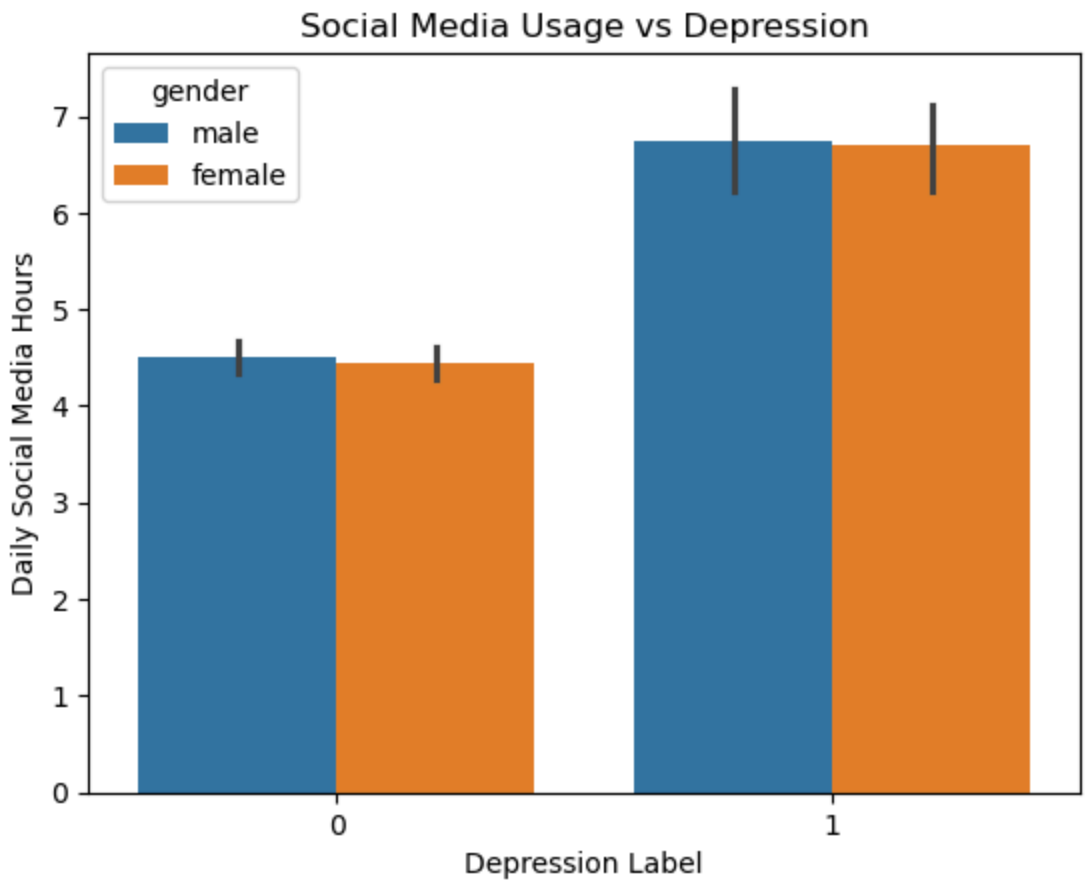
```
sns.barplot(x='platform_usage', y='depression_label' ,hue = 'gender', data=df)
plt.title('Depression rate of  Male and Female by Platform Usage')
plt.xlabel('Social Media Platform')
plt.ylabel('Depression Rate')
plt.show()
```



Social Media Usage Vs Depression Label

In [151…

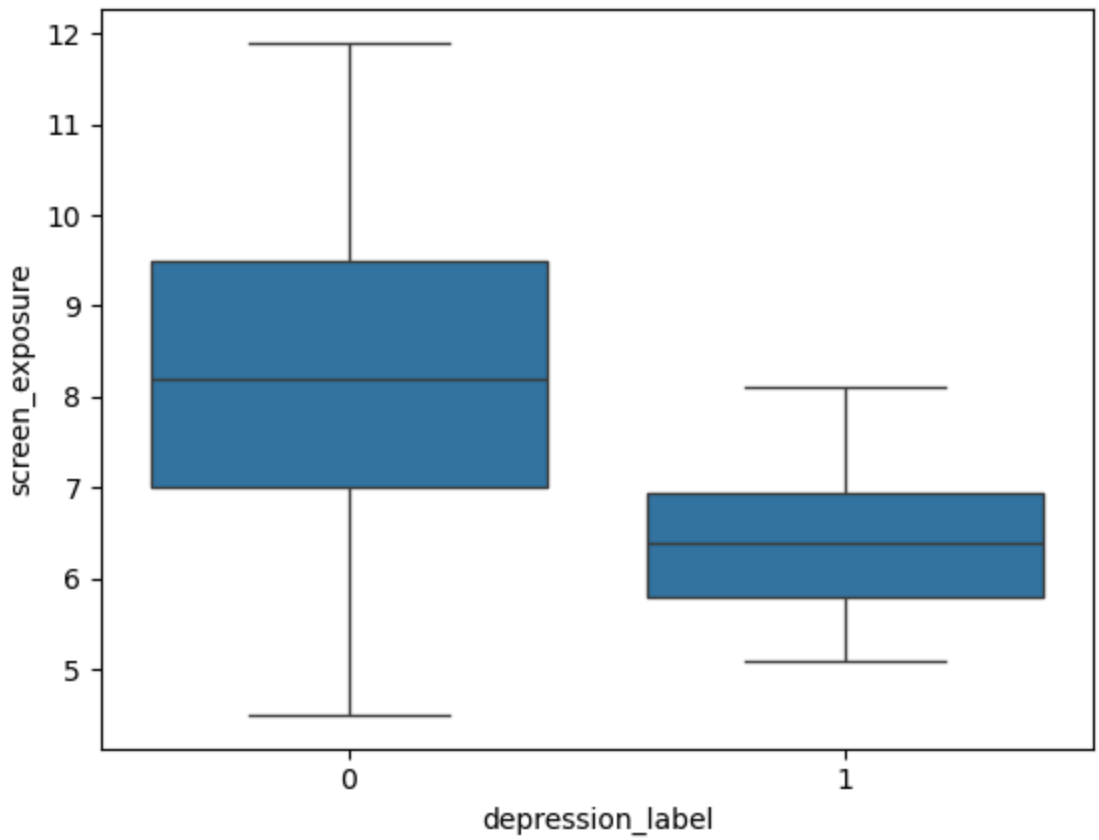
```
sns.barplot(x='depression_label', y='daily_social_media_hours',hue='gender', d
plt.title('Social Media Usage vs Depression')
plt.xlabel('Depression Label')
plt.ylabel('Daily Social Media Hours')
plt.show()
```



Depression Label vs Screen Exposure

In [163…

```
sns.boxplot(x='depression_label',y='screen_exposure',data=df)
plt.show()
```

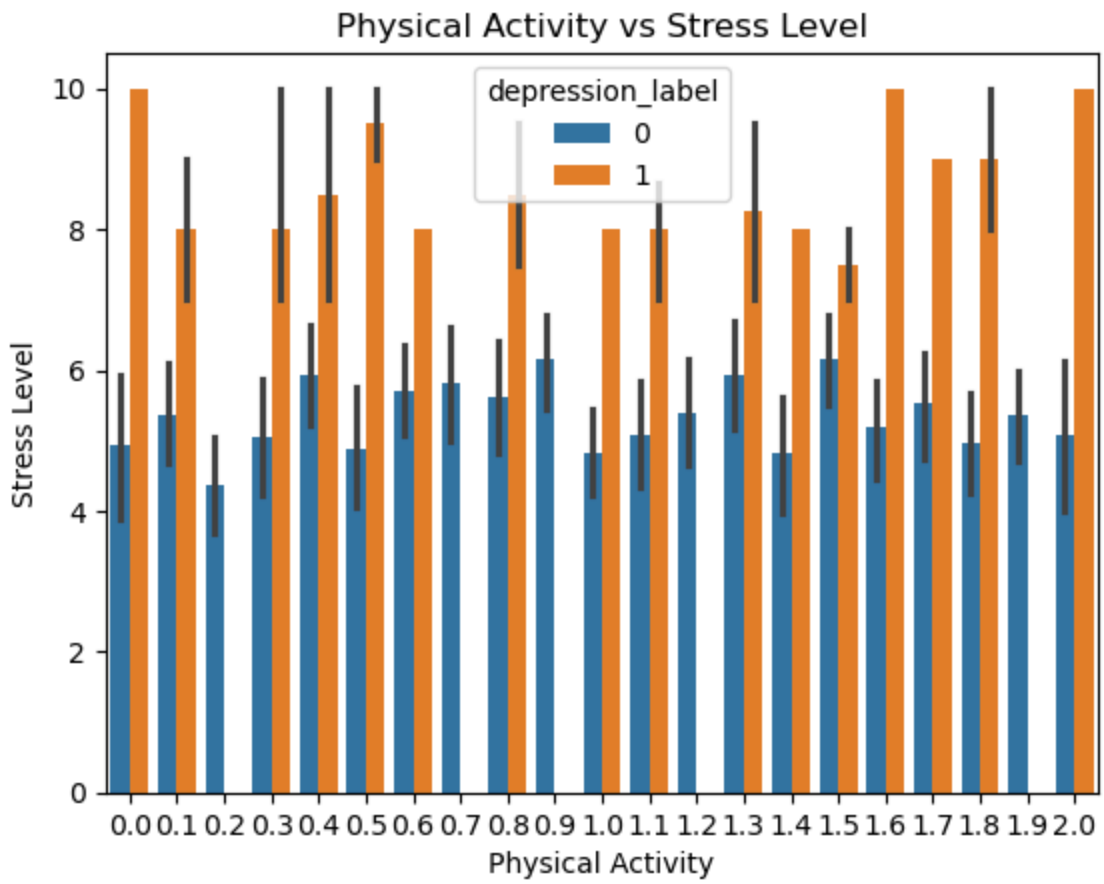


In [17]:
```
df.columns
```

Out[17]:
```
Index(['age', 'gender', 'daily_social_media_hours', 'platform_usage',
       'sleep_hours', 'screen_time_before_sleep', 'academic_performance',
       'physical_activity', 'social_interaction_level', 'stress_level',
       'anxiety_level', 'addiction_level', 'depression_label',
       'screen_exposure'],
      dtype='object')
```

Physical Activity vs Stress Level

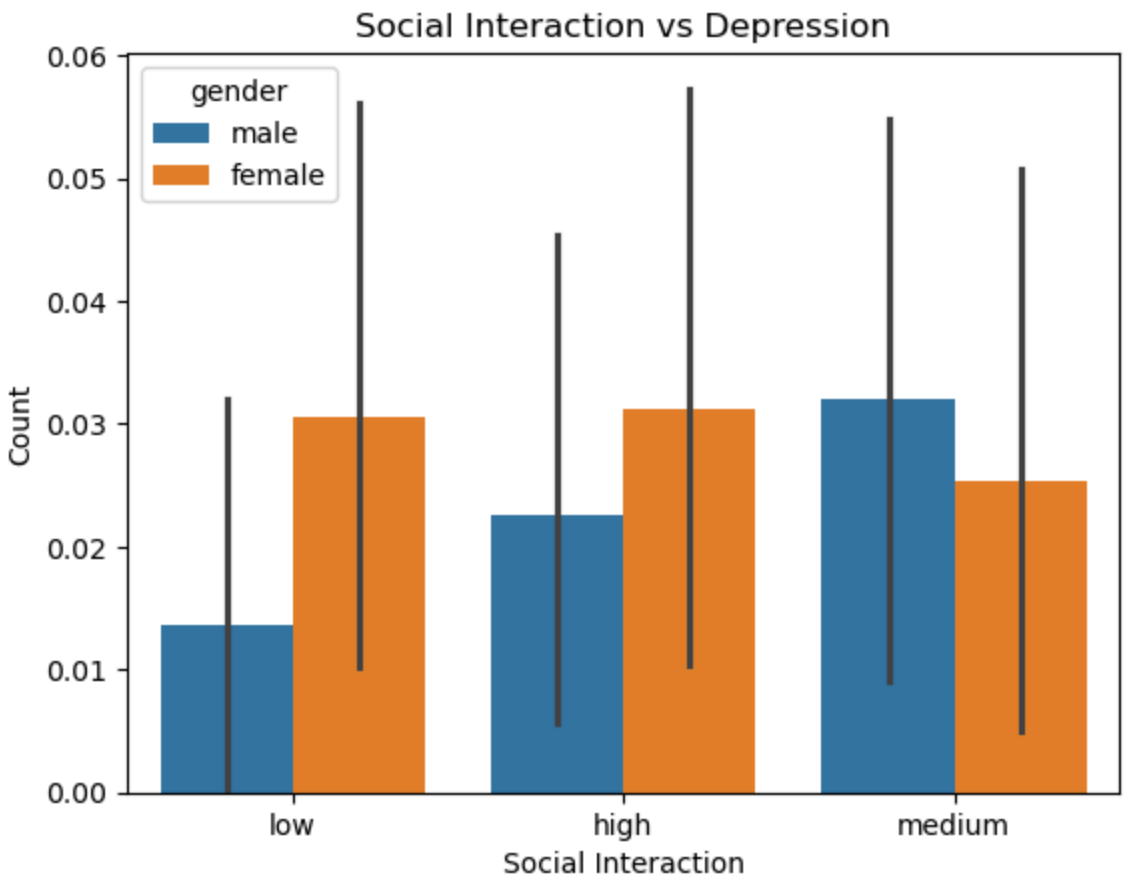
In [19]:
```
sns.barplot(x='physical_activity',y='stress_level',hue='depression_label',data
plt.xlabel('Physical Activity')
plt.ylabel('Stress Level')
plt.title('Physical Activity vs Stress Level')
plt.show()
```



Social Interaction vs Depression

In [145…

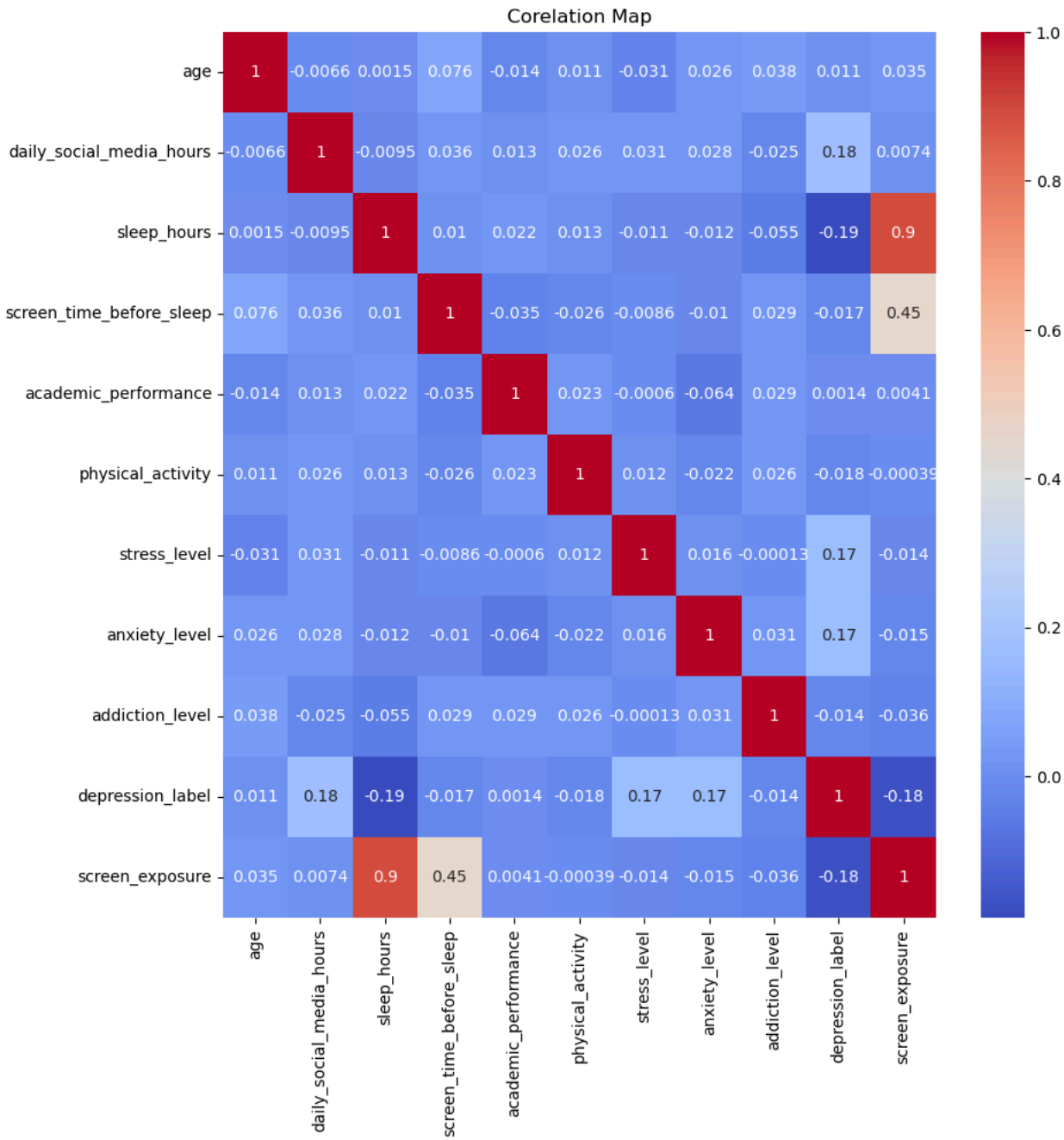
```
sns.barplot(x='social_interaction_level', y='depression_label',hue='gender', d
plt.title('Social Interaction vs Depression')
plt.xlabel('Social Interaction')
plt.ylabel('Count')
plt.show()
```



Corelation Map

In [103…

```
plt.figure(figsize=(10,10))
sns.heatmap(df.corr(numeric_only=True) , annot = True , cmap='coolwarm')
plt.title('Corelation Map')
plt.show()
```



In [121…

```
df['gender'].unique()
df['platform_usage'].unique()
#df['social_interaction_level'].unique
```

Out[121…

```
array(['Instagram', 'TikTok', 'Both'], dtype=object)
```

Feature Engineering

In [51]:

```
df['gender'] = df['gender'].map({'female':0 , 'male':1})
```

In [53]:

```
df['platform_usage'] = df['platform_usage'].map({'Instagram':0 , 'TikTok':1 ,
```

In [177…
```
df['social_interaction_level'] = df['social_interaction_level'].map({'low':0 ,
```

Model Training Using ML Algorithms

In [63]:
```
x = df.drop(columns='depression_label',axis=1)
y = df['depression_label']
```

In [65]:
```
from sklearn.model_selection import train_test_split
```

In [67]:
```
x_train , x_test , y_train , y_test = train_test_split(x,y,test_size=0.2,randc
```

Using Algorithms For Training

1. Logistic Regression

2. RandomForestClassifier

3. XGBClassifier

4. DecisionTreeClassifier

In [69]:
```
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from xgboost import XGBClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import classification_report , accuracy_score
```

Logistic Regression

In [73]:
```
lr = LogisticRegression()
lr.fit(x_train,y_train)
```

Out[73]:
```
▾ LogisticRegression
LogisticRegression()
```

In [77]:
```
y_pred_lr = lr.predict(x_test)
accu_lr = accuracy_score(y_test , y_pred_lr)
```

In [87]:
```
print('Logistic Regression Accuracy :' , accu_lr)
```

```
Logistic Regression Accuracy : 0.9916666666666667
```

In [81]:
```
print(classification_report(y_test, y_pred_lr))
```

```
              precision    recall  f1-score   support

           0       0.99      1.00      1.00       234
           1       1.00      0.67      0.80         6

    accuracy                           0.99       240
   macro avg       1.00      0.83      0.90       240
weighted avg       0.99      0.99      0.99       240
```

RandomForestClassifier

In [95]:
```
rfc = RandomForestClassifier(random_state=42)
rfc.fit(x_train,y_train)
```

Out[95]:
```
▾ RandomForestClassifier
RandomForestClassifier(random_state=42)
```

In [97]:
```
y_pred_rfc = rfc.predict(x_test)
accu_rfc = accuracy_score(y_test , y_pred_rfc)
```

In [101…
```
print('RandomForestClassifier Accuracy :', accu_rfc)
```

```
RandomForestClassifier Accuracy : 0.9791666666666666
```

In [103…
```
print(classification_report(y_test, y_pred_rfc))
```

```
              precision    recall  f1-score   support

           0       0.98      1.00      0.99       234
           1       1.00      0.17      0.29         6

    accuracy                           0.98       240
   macro avg       0.99      0.58      0.64       240
weighted avg       0.98      0.98      0.97       240
```

XGBClassifier

In [105…
```
xg = XGBClassifier()
xg.fit(x_train,y_train)
```

Out[105…

```
XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric=None, feature_types=None,
              feature_weights=None, gamma=None, grow_policy=None,
              importance_type=None, interaction_constraints=None,
              learning_rate=None, max_bin=None, max_cat_threshold=None,
```

In [107…

```
y_pred_xg = xg.predict(x_test)
accu_xg = accuracy_score(y_test , y_pred_xg)
```

In [109…

```
print('XGBClassifier Accuracy :',accu_xg)
```

```
XGBClassifier Accuracy : 0.9958333333333333
```

In [111…

```
print(classification_report(y_test , y_pred_xg))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       234
           1       1.00      0.83      0.91         6

    accuracy                           1.00       240
   macro avg       1.00      0.92      0.95       240
weighted avg       1.00      1.00      1.00       240
```

DecisionTreeClassifier

In [113…

```
dt = DecisionTreeClassifier(random_state=42)
dt.fit(x_train,y_train)
```

Out[113…

```
DecisionTreeClassifier
DecisionTreeClassifier(random_state=42)
```

In [115…

```
y_pred_dt = dt.predict(x_test)
accu_dt = accuracy_score(y_test , y_pred_dt)
```

In [117…

```
print("Decision Tree Classifier Accuracy :", accu_dt)
```

```
Decision Tree Classifier Accuracy : 0.9916666666666667
```

In [119…

```
print(classification_report(y_test , y_pred_dt))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       234
           1       0.83      0.83      0.83         6

    accuracy                           0.99       240
   macro avg       0.91      0.91      0.91       240
weighted avg       0.99      0.99      0.99       240
```

Summary

Logistic Regression Accuracy : 0.9916

RandomForestClassifier Accuracy : 0.9791

XGBClassifier Accuracy : 0.9958

Decision Tree Classifier Accuracy : 0.9916

In [ ]:

```
##
```